K.M.S

COPYRIGHT RESERVED

BCA (Sem-III) – Software Testing (304) SEC-I

2025

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer all group as directed.

Group—A

(Objective Type Questions)

1. Choose the correct answer : $2 \times 10 = 20$

(i) Which of the following is the primary objective of software testing?

a) To increase development cost

b) To identify defects and ensure software quality

c) To improve hardware efficiency

d) To replace debugging activities

(ii) In the V-Shaped Model, test planning activities begin during:

a) Coding phase

b) Maintenance phase

c) Corresponding development phases

d) Deployment phase

(iii) Which black-box testing technique divides input data into valid and invalid partitions?

a) Basis Path Testing

b) Decision Table Testing

c) Equivalence Class Testing

d) Statement Testing

(iv) Boundary Value Analysis is most effective in identifying defects related to:

a) Database design

b) Input limits and ranges

c) User interface layout

d) System documentation

(v) Decision Table Testing is particularly useful when:

a) The software contains loops

b) Internal code structure needs examination

c) Complex business rules exist with multiple conditions

d) Performance testing is required

(vi) In white-box testing, Cyclomatic Complexity helps determine:

a) Number of defects in software

b) Number of independent execution paths

c) Execution time of the program

d) Memory utilization

(vii) Which coverage criterion ensures that every executable statement is executed at least once?

a) Path Coverage

b) Branch Coverage

c) Statement Coverage

d) Condition Coverage

(viii) A DD Path Graph is mainly used in:

a) Equivalence Partitioning

b) Basis Path Testing

c) Regression Testing

d) System Testing

(ix) Which of the following provides the highest level of structural testing coverage?

a) Statement Coverage

b) Branch Coverage

c) Condition Coverage

d) Path Coverage

(x) Validation testing answers which of the following questions?

a) Are we building the product right?

b) Are we building the right product?

c) Is the code optimized?

d) Is the documentation complete?

Group – B

(Short Answer Type Questions)

Answer any **four** of the following : 5×4=20

2. Define software testing strategy. Explain its significance in software development.

3. Distinguish between Verification and Validation with suitable examples.

4. Explain the concept of Equivalence Class Testing with an appropriate example.

5. What are the advantages and limitations of Boundary Value Analysis?
6. Define Program Graph and discuss its importance in Basis Path Testing.
7. Differentiate between Statement Coverage and Branch Coverage.
8. What is Decision Table Testing? Illustrate its application with a simple example.

Group – C

(Long Answer Type Questions)

Answer any **two** of the following : 15×2=30

9. Describe the V-Shaped Software Development Lifecycle Model. Discuss its advantages and disadvantages with a neat diagram.
10. Discuss the Strategic Approach to Software Testing in detail. Explain the various levels of testing involved in conventional software development and their objectives.

11. Explain Validation Testing and System Testing. Compare both testing approaches with suitable examples and discuss their importance in ensuring software quality.
12. Consider a university admission system where eligibility depends on the following conditions:
    Candidate must score at least 50% marks.
    Reserved category candidates receive a 5% relaxation.
    Age should not exceed 25 years.
    Construct a Decision Table for the above scenario and derive suitable test cases.

———×———